

Empirical Proof Record: VALIDATED

2026-05-21T22:54:01.854090+00:00

Empirical Proof Record — VALIDATED

Proof ID: 70df74482bac808c90697bdb9aaea00d...

Created: 2026-05-21T22:54:01.854090+00:00

1. Claim

Across a trajectory with spaced re-exposures, the substrate keeps recognising every re-exposed stimulus: (for every same-stimulus re-exposure pair (i, j) in the trajectory, $\text{Jaccard}(\text{concepts_i}, \text{concepts_j}) \geq 0.80$). The reported floor is the worst pair across the whole run.

- **Operationalisation:** Run a schedule of $\text{len}(\text{stimuli})$ unique stimuli each repeated $3\times$ and interleaved ($\text{gap} = \text{len}(\text{stimuli})$) through Kimera’s entity target (Takwin). For each stimulus, take the per-cycle “concepts” set at each exposure and compute Jaccard over all exposure pairs. $\text{recognition_jaccard_floor} = \min$ over all pairs across all stimuli; the LTL invariant holds iff $\text{floor} \geq \theta$.
- **Threshold:** $\text{recognition_jaccard_floor} \geq 0.8 \text{ jaccard}$
- **H0:** $\text{H0: floor} < 0.80$ — recognition collapses under manifold deformation somewhere in the run
- **H1:** $\text{H1: floor} \geq 0.80$ — recognition is stable across the whole trajectory (memory-as-deformation holds as a flow property)

2. Verdict

VALIDATED — observed 0.8636363636363636

recognition floor 0.8636 over 24 re-exposure pairs ($8 \text{ stimuli} \times 3 \text{ exposures}$); mean Jaccard 0.9629; 0 exposures produced no concept set; worst pair: stimulus 3 cycles $3 \leftrightarrow 11 = 0.8636$

3. Evidence

Pillar	Statistic	Value	95% CI	Lib
0.flow.recognition_stability	recognition_jaccard_floor	0.8636363636363636	—	oph

4. Pre-registration

- Registered at: 2026-05-21T22:54:01.493608+00:00
- Config hash: 0fbe8ce56b2a2db7203910b72aaebc0c...
- Data hash: 0289f594eb2ea25b582960992cf2db41...

5. Signature

b3a25e50888df1ce75f4f81e12f1ab3cac5c83926a5fb7c8d0929e49c793ceea